Learning Airline Route Preferences from Flight-Plan Revisions to Support Fuel-Efficient Alternatives

Ramon Dalmau[1], David Perez[2], Seddik Belkoura[2], Franck Ballerini[1],
Vincent Taverniers[2], Gratiel Marius Marin[2], Robin Deransy[1], Benjamin Cramet[3] and Grigori Gustin[3]
EUROCONTROL
[1]Innovation Hub, Brétigny-sur-Orge, France [2]Maastricht UAC, the Netherlands [3]Headquarters, Brussels, Belgium

***Abstract*—Flight-plan revisions provide a direct record of route choice: an airline replaces a route it previously filed with another route for the same flight. We use 1.48 million such pairs from eighteen months of European traffic to learn which of two routes the airline is likely to choose. The contribution is a way to build and evaluate preference models from these recorded comparisons, without assuming that an unfiled route was considered and rejected. The model combines route indicators, flight context and waypoint identifiers, while preventing filing times from revealing which route came later. On a later, held-out quarter, it identifies the chosen route in 68.1 % of cases, compared with 55.0 % for the best single-indicator rule. Accuracy reaches 94.9 % on the fifth of pairs with the highest confidence. A smaller set of flights that kept their routes, matched to similar flights by day, origin, destination and aircraft type, helps the model represent that behaviour as well. To illustrate how the predictions could guide fuel-saving proposals, we evaluate a policy that selects the lower-fuel route only above a calibrated confidence threshold. Of its selected routes, 79.2 % match the airline's subsequent filing. These matches account for 20.1 kilotonnes of planned-fuel differences over the quarter. The results demonstrate useful prediction of recorded route choices; acceptance of real proposals and additional fuel savings remain questions for prospective evaluation.**

***Keywords*—air traffic flow management, airline preferences, learning to rank, fuel efficiency**

I. Introduction

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the operator subsequently re-filed. The new route shed that delay completely, and it cost the airline on each of the three counts it pays for directly: 39 minutes of flight time, 1 397 kg of extra planned fuel, and 466 euros more in en-route charges. The records therefore show a substantial trade-off between attributed delay and the other route indicators. They do not reveal the operator's complete reasoning or its internal costs.

This paper asks whether such recorded choices can help predict route preferences. Airlines revise flight plans for several reasons: to respond to delay, updated weather, changes in an aircraft's earlier flights, or route availability. Whatever prompted the change, the archive contains the route replaced and its replacement. We use those two routes as a training pair and ask the model to identify the replacement from their characteristics, without revealing the filing order.

The central methodological choice is how to obtain the training labels. A common approach ranks a filed route above alternatives generated later. Those alternatives may be feasible, but the record does not establish whether the airline considered them. A revision instead supplies two routes that the airline filed for the same flight. It is direct evidence of a change in filing, although changing constraints mean that it is not always a comparison between two options still available at the same time.

The contribution has three parts. First, we construct route comparisons from revisions while removing information that would reveal the answer merely through message timing. Secondly, we evaluate both prediction accuracy and confidence on a later period, and use matched flights that kept their routes to address the bias towards changes. Thirdly, we examine which inputs support the predictions and illustrate selective use of the model in a retrospective fuel-saving proposal policy. The scientific question is whether these records contain useful, predictable information about route choice. The proposal policy provides one way to examine the value of that information.

The remainder of this manuscript is organised as follows. Section II explains why a pair of routes drawn from a revision constitutes a preference observation and what a model of such observations is for, and Section III positions the work within the literature on rerouting and route choice. Section IV describes the archive, the split protocol and the model, while Section V reports the results. Finally, Section VI concludes and outlines future work.

II. Background

This section explains how a flight-plan revision becomes a labelled preference pair, why that label is preferred to one built from generated candidates, and what the model is for.

A. Flight-plan revisions as preference data

In the European network a flight plan is filed before departure and may be revised until the flight leaves. Every filing is a message, and so is every later change. The network keeps them all, so what it holds for a flight is the whole sequence of messages sent for it. A route change is accordingly a pair of consecutive messages rather than a record of its own: the message that changed the route, and the one it replaced.

The two together describe a decision, because the operator had committed to the earlier route and then filed the later one for the same flight, which makes the later route the one it chose and the earlier one the route it abandoned.

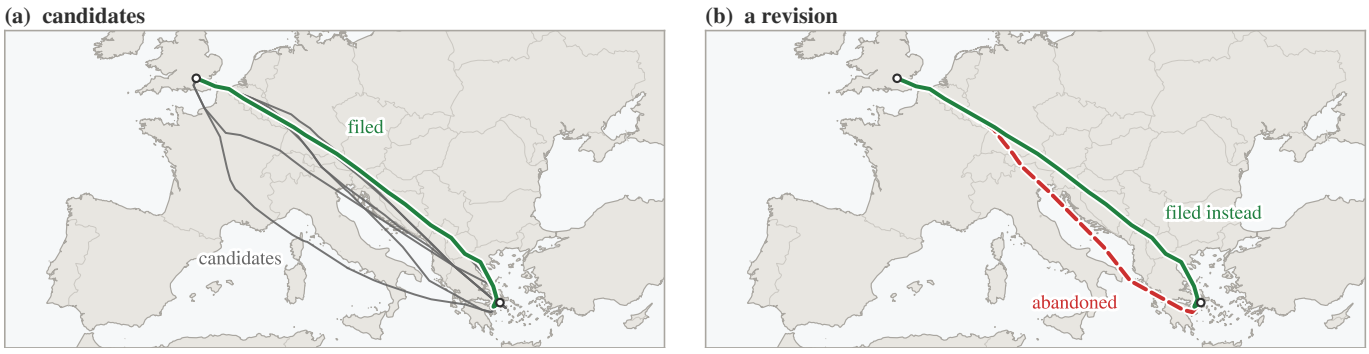


Figure 1. Two ways to build a training pair, on one city pair. Left: the filed route (solid green) is ranked above candidates a route generator would supply (grey); the grey tracks drawn here are real routes flown on the same city pair, standing in for generated ones. Right: a revision supplies two routes that the same operator filed for the same flight hours apart, one of which it abandoned. The right-hand pair directly records which filed route replaced the other.

The sharpest of those decisions are made under flow management. When the traffic expected in a volume of airspace exceeds what can safely be handled there, whether because of demand, staffing, weather or equipment, a *regulation* is placed on it and the flights crossing it are given calculated take-off times at the rate it can absorb; the difference between a flight's calculated and requested take-off times is its ATFM delay.

An operator facing an ATFM delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel and en-route charges, and neither answer is universally correct. For instance, a flight with a tight aircraft rotation, in which one aircraft operates several successive legs, may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. This is why the decision is worth learning rather than deriving: the costs that determine it are operator-specific and flight-specific, and the network does not observe them, though the outcomes of those trades are recorded. Published European reference values [1] give the order of magnitude of the cost of delay and are widely used for analysis, but they are not the cost function any particular operator applies.

Seven pairs in ten in this archive carry no regulation on either route, and within those the two routes differ by a median of one kilogramme of planned fuel; the full breakdown by decision context is in Section IV.

Using choices to learn preferences follows the idea of revealed preference [2]. Here the observation is a replacement: route B was filed after route A for the same flight. We call B the *chosen* route and A the *abandoned* route throughout the paper. These terms refer to their roles in the record, not to the scores later assigned by the model.

An alternative construction, used in the authors' earlier work [3], ranks a filed plan above routes supplied by a generator. Those routes may be feasible, but the record does not show whether the airline considered them. Labelling each generated route as less preferred therefore makes an assumption beyond the observed filing.

A revision pair provides a more direct comparison (Fig. 1): the operator filed both routes in succession for the same flight. The label records which route replaced the other. It does not establish that a dispatcher manually compared them, or that the earlier route remained available. We therefore use the recorded choice as a proxy for preference, rather than as a measurement of the airline's underlying utility.

With generated alternatives, a model may partly learn to distinguish filed routes from generated ones. Revision pairs reduce that ambiguity because both routes passed through the airline's filing process. This is a reason for the label construction, not a measured comparison of the two training approaches on identical flights.

B. Potential uses of the predictions

One potential use is to screen fuel-saving route proposals. A model first ranks two routes; calibration then estimates how often its higher-ranked route matches the recorded choice. A threshold can retain only predictions with sufficient confidence. Raising it usually reduces the number of proposals and increases their precision, defined here as the fraction that match the airline's subsequent filing. A target precision is selected on earlier data and measured on later data; it is not guaranteed by the threshold.

A second research question is whether predicted route preferences could help prioritise voluntary rerouting requests when traffic demand exceeds available capacity. Studying that question would require a joint model of route acceptability and the effect of each reroute on demand.

Here we evaluate the preference model and simulate the first use with historical revisions. Actual proposal acceptance and overload release remain outside this evaluation.

III. Related Work

Work on rerouting in ATFM has largely approached the problem from the network side. Over two decades ago, the authors of [4] formulated rerouting as a dynamic network flow problem, and the authors of [5] subsequently added a stochastic model that reroutes flights as capacity evolves. Both optimise an objective defined by the network and assume, in effect, that the resulting assignment is executed, so that the willingness of the airline to fly it is not modelled.

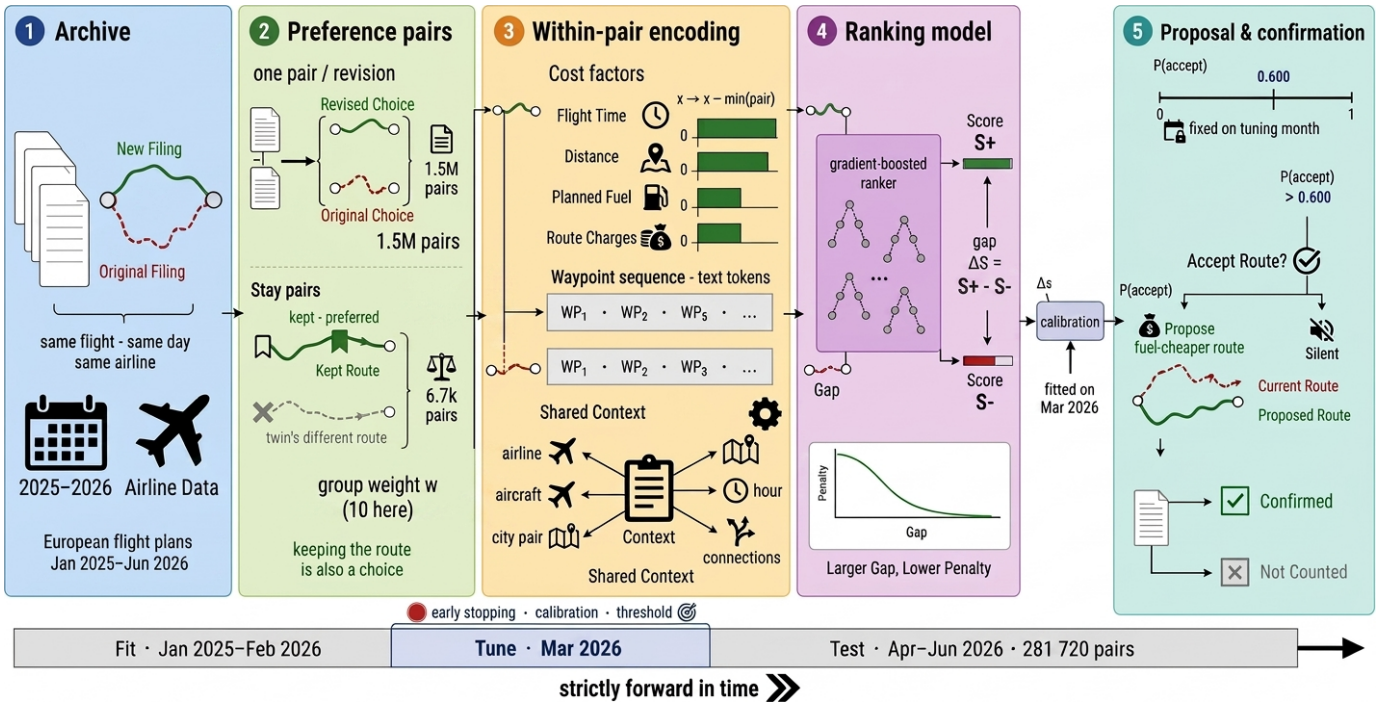


Figure 2. Workflow, from filing records to retrospective proposals. The score bars illustrate a correctly ranked pair; calibration uses the size of the score gap. Here $P(\text{accept})$ is shorthand for the estimated probability that the candidate matches the recorded choice, not a measured response to a proposal. "Confirmed" means a match with the later filing. Stay-pair preferences are inferred from matched flights. The fit, tune and test periods have separate roles.

More recently, route choice in Europe was studied as a trade between fuel and en-route charges [6], and as a discrete-choice model of the same trade [7], which documents an economic aspect of route choice, but uses no label in which the operator rejected an alternative it had itself filed. The closest work to the present one is that of the authors of [8] and [9], who incorporate airline preferences and trajectory options into collaborative rerouting and show that accounting for them changes which reroutes are selected. Those preferences, however, are inputs to an optimisation, whereas this paper learns the preference function itself, at network scale, from what airlines did rather than from what they stated.

The direct predecessor of this work applies learning to rank [10] to flight routes for a different purpose: predicting the flight plans that have not yet been filed, so that traffic demand can be estimated earlier than the filing horizon allows [3].

That work established the framing adopted herein, namely a gradient-boosted ranker that scores routes by their indicators in context, and it reported the very difficulty that motivates the present paper. Its training pairs rank filed plans above generated alternatives. Section II explains how revision pairs differ.

Machine learning has also been used to forecast the routes and flight levels an airline will request [11], [12]. The acceptance of the reroutes it is offered has been learned from its past requests [13], [14] and demonstrated operationally [15]. The route structure such choices act on has been defined and visualised from traffic [16], [17], and the cost of confining optimisation to published waypoints measured [18]: each predicts or prescribes routes from outside the airline, leaving its preference unlearned.

IV. Methodology

Figure 2 separates the five steps of the method. Historical filings supply labels during training and evaluation. When the model scores a pair, it receives route characteristics and shared flight context, not the chosen/abandoned labels. The comparison with the label is made after scoring. In the diagram's final panel, the probability is always that of the lower-fuel candidate: if the model ranks that route second, we use one minus the probability of its first-ranked route.

A. Data and pair construction

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs. A pair is one training example with two rows, one per route. Its label identifies which row is the replacement; the model must infer that label from the inputs. Throughout the paper, *planned fuel* is the fuel estimate attached to the recorded plan, not a measurement taken after the flight. The four route indicators (flight time, distance, planned fuel and route charges) are all such estimates, and the model reads them exactly as the plan filed them: it does not recompute a route's flight time against the winds of the day, nor does it take the day's weather as a separate input. The indicators therefore reflect the conditions the plan assumed when it was filed, and weather enters a comparison only indirectly, through the plan it already shaped.

TABLE I. THE REVISION ARCHIVE BY DECISION CONTEXT, OVER ALL EIGHTEEN MONTHS. THE FUEL DIFFERENCE IS BETWEEN THE TWO ROUTES OF A PAIR.

context	pairs	share [%]	median Δfuel (new − old) [kg]	new route burns less [%]
unregulated	1 047 228	70.6	1	46.2
left regulation	117 594	7.9	101	28.1
reduced delay	155 326	10.5	65	31.6
kept or raised delay	141 648	9.6	0	47.4
became regulated	21 337	1.4	−5	51.8

The model learns which of two recorded plans an airline kept, given the plans as filed, not what the weather that day would have favoured.

What these revisions look like matters before any model touches them, and Table I breaks the archive down by the regulation context in which each revision was made.

A route counts as regulated when the flight filed on it carried a regulation, even at zero delay, so the pairs fall by where the regulation sat: on neither route, on the abandoned route only, on both, or on the newly filed route only. Most pairs have no regulation on either route, and their median planned-fuel difference is one kilogramme. Of the regulated pairs, two groups together account for 18.4 %: those leaving a regulation (the old route was regulated, the new one is not) and those reducing delay while both routes stay regulated. Their median planned-fuel increases are 101 and 65 kg, yet only 28.1 % and 31.6 % of their replacements burn less fuel than the route they replaced. The remaining regulated pairs either kept or raised the delay on both routes, or became regulated, the new route alone now carrying a regulation.

Each route is described by twenty-three features, summarised in Table II. The two routes of a pair are not the same plan twice over: they are two plans for one flight, filed weeks apart, the second replacing the first. Because the replacement is always the later filing, any feature that drifts with time would identify the answer without the model assessing the route at all. Time to departure is the clearest case: it is always smaller for the later filing, so a model given it in raw form could answer "the one filed more recently", be right on nearly every pair, and learn nothing about routes. A high accuracy would hide this: the model would look skilled while merely reading the filing dates.

To make the two routes comparable, the features fall into two kinds. Route-level features (the four indicators, the regulation fields and the waypoint tokens) differ between the routes of a pair, and the model uses their differences. Flight-level features (the operator, aircraft type, city pair, connection times and time to departure) belong to the flight and, apart from the outbound connection time, carry the same value on both routes. They are harmonised by taking each one's value from the later filing; because the replacement is always the later filing, both routes then carry the smaller time to departure. No flight-level feature can separate the two routes, which closes the timing shortcut above: the model can no longer answer by reading which plan was filed more recently.

TABLE II. WHAT THE MODEL SEES FOR EACH ROUTE; SOME ROWS SUMMARISE SEVERAL FEATURES. THE ROUTE-LEVEL GROUPS DIFFER WITHIN A PAIR; THE FLIGHT-LEVEL GROUPS ARE HARMONISED ACROSS THE PAIR, APART FROM THE OUTBOUND CONNECTION TIME.

group	content
indicators	flight time, distance, planned fuel, route charges
regulation	attributed delay (own, inbound, outbound), their count, reason, and the type and identifier of the protected volume, meaning the airspace whose capacity the regulation protects
rotation	inbound and outbound connection times and states
context	operator, aircraft type, city pair, hour, weekday, month, time to departure
route	waypoint identifiers, encoded as unordered tokens

TABLE III. ILLUSTRATIVE INPUT PREPARATION FOR A HYPOTHETICAL REVISION: RAW VALUE → MODEL INPUT. THE VALUES ARE ILLUSTRATIVE, NOT STUDY RESULTS.

input	route A	route B
planned fuel [kg]	2 000 → 0	2 300 → 300
flight time [min]	60 → 0	65 → 5
distance [km]	400 → 0	410 → 10
route charges [EUR]	100 → 20	80 → 0
time to departure [min]	120 → 90	90 → 90

The route-level indicators are then expressed relative to the smaller value in each pair, so the model sees how much more one option requires than its absolute level. Planned fuel of 2 000 and 2 300 kg becomes 0 and 300 kg; a zero means "no extra fuel relative to the other route", not a flight using no fuel. Each indicator is normalised separately: in Table III, route A needs less fuel than route B yet pays higher route charges, so neither route is cheapest on every indicator. Harmonising the flight context removes a timing shortcut and normalising the route indicators changes their representation: two separate operations, and the table works one revision through both, turning the raw fuel and charge values into comparisons while time to departure takes the later filing's value on both rows.

Waypoint identifiers provide a second description of each route. They are treated as words in an unordered collection, so the model knows which published points a route uses but not their order or coordinates. Two routes can have similar fuel and distance indicators yet pass through different parts of the network. These identifiers allow the model to learn associations that the aggregate indicators alone cannot express.

Revision pairs describe flights that changed route. They omit flights that retained a route despite a delay, so training on revisions alone may favour alternatives too readily. We add a smaller set of *stay pairs* to represent the latter behaviour.

A stay pair joins two flights on the same day, with the same origin, destination and aircraft type. Neither revised its route; one was regulated and the other was not. The regulated flight's retained route is labelled preferred, and the other flight's route supplies a comparison. Unlike a revision pair, this is an inferred label: the record does not show that the regulated flight considered or could have used that alternative at its departure time. Both rows are evaluated with the retained-route flight's context. The comparison therefore asks which route suits that flight; it does not rank the two flights against each other.

There are 6 672 stay pairs, including 1 093 in the test months. They are split by time and added to the corresponding revision subsets. During training, each stay pair carries a larger group weight, so it contributes more to the ranking loss than a revision pair does. Section V-B reports the effect on both revision and stay-pair accuracy.

The time split separates three tasks. The fitting period teaches the model how features relate to recorded choices. The tuning period selects when to stop training, fits the probability mapping and sets the policy threshold. The final test period measures how those fixed choices perform on later data. Using different periods matters because a rule that works on the records used to choose it may perform differently afterwards.

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026 (1 134 064 pairs), tuned on March 2026 (67 349), and evaluated once on April to June 2026 (281 720). Every threshold, calibrator and stopping decision is taken on the tuning month alone, and flights appearing in the tuning or evaluation months are removed from the fitting set.

All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements. Operator identity enters the model as a categorical feature only, and no operator or airspace volume is named anywhere in this paper. The unit of observation is a route pair; a flight may contribute more than one revision. Pair counts should therefore not be read as counts of distinct flights.

B. Model, training and calibration

The operational question is comparative: given two routes for one flight, which of them does the airline prefer? Accordingly, the model gives each route a single number, which is not a probability and has no units, and which is only meaningful next to the number given to another route for the same flight. Training pushes those numbers apart in the right direction. For each labelled pair, a penalty increases when the less-preferred route scores higher. It falls as the preferred route's lead grows, encouraging the correct ordering and a larger gap even once the order is right [19]. Nothing in the objective asks the model to reproduce a route in isolation, which is what makes the pair the natural unit.

The scoring function is a CatBoost ensemble of gradient-boosted decision trees [20], which combines many simple trees so that each adds a small correction to the running score. Shared flight attributes still matter even though they cannot distinguish the two routes on their own: they can change how the model interprets a route difference. The trees capture these interactions, such as a flight-time difference having a different effect when little time remains before the aircraft's next leg. Tree depth is 8, with the learning rate selected automatically. Early stopping uses the tuning-period criterion to select how many trees to retain, rather than continuing training indefinitely. Stay pairs enter the fitting set with a group weight w, swept over $\{10, 30, 100\}$; Section V-B reports the effect of the adopted weight 10.

The scores indicate an ordering, not a probability. For example, scores of -2.82 and $+0.90$ favour the second route by about 3.7 score units. Calibration maps the size of this gap to an estimated probability that the higher-scoring route matches the recorded choice. A probability of 0.8 means that, among comparable predictions, about eight in ten are expected to be correct. For the example's gap of 3.7, the fitted mapping returns 0.99.

To fit this mapping, the trained model scores pairs from the tuning month. Their known labels reveal whether each ordering is correct. Calibration relates the observed correctness to the gap: pairs with similar gaps supply evidence about how reliable such predictions are. The resulting mapping is then fixed before the test quarter. This step changes the probability attached to an ordering; it does not retrain the route-scoring model.

We compare three mappings on the tuning month: the raw sigmoid of the gap, a fitted sigmoid (Platt scaling), and a monotone stepwise mapping (isotonic regression) [21], [22]. Isotonic regression is constrained not to decrease as the gap grows. It has the lowest tuning-month log loss, a measure that penalises confident errors, and is retained for testing. On the test quarter, the average discrepancy between predicted and observed rates is 1.8 percentage points for that mapping and 0.9 for the raw sigmoid. These averages describe calibration across probability bands; they do not bound the error of every prediction. The test result is reported without changing the earlier selection.

The retrospective proposal policy uses this probability in three steps. First, exclude pairs with equal planned fuel; 97.9 % remain eligible. Secondly, identify the lower-fuel route. If it is the higher-scoring route, use the calibrated probability; otherwise use one minus that probability. Thirdly, select it only if that probability exceeds τ. The threshold $\tau = 0.600$ was chosen on the tuning month to target 80 % precision and is applied unchanged to the test quarter. A 0.600 probability threshold and an 80 % precision target are different quantities: the latter is the average correctness of all proposals above the former.

Consider two hypothetical pairs. In the first, the lower-fuel route has calibrated probability 0.85 of matching the later filing; it passes the threshold. In the second, the lower-fuel route has probability 0.40; it is withheld even if its fuel difference is large. The policy therefore requires both lower planned fuel and sufficient predicted preference. These illustrative probabilities are not additional experimental results.

This policy compares the two routes without using which was filed first. It may therefore select either the replacement or the route that was abandoned. A selection is counted as confirmed only when it matches the replacement. This is a retrospective test of choice prediction, using an alternative known from the historical record.

V. Results

This section reports how accurately the model predicts airline choices, whether the change bias of Section IV is

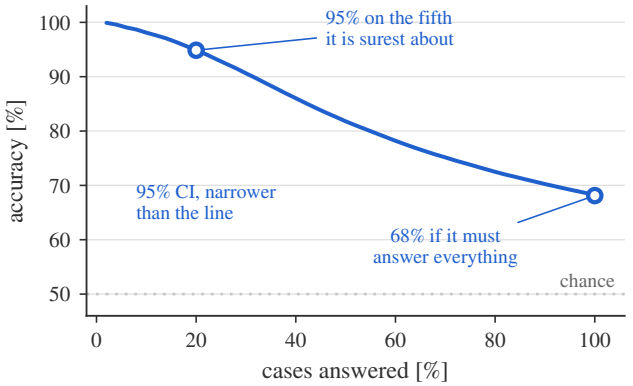


Figure 3. Accuracy against the share of cases answered, on the held-out quarter, a case being one pair of routes. Moving left, the model answers fewer cases and is right more often. The least confident predictions are dropped first; the dotted line marks random choice.

controlled, what the model has learned about route preference, and what the retrospective proposal policy selects.

A. Predictive performance

On the held-out quarter (April to June 2026, 281 720 pairs) the model identifies the airline's chosen route 68.1 % of the time (95 % interval 67.9–68.3), against 50 % for random choice and 55.0 % for the best single-indicator rule.

No tested single-indicator rule predicts much of the choice. The delay rule reaches 71.6 % on the 14.6 % of pairs where both delay values are available and unequal; the remaining 85.4 % fall outside the rule and are scored as a random choice. Weighting the two groups gives 53.2 % overall ($0.146 \times 71.6\,\% + 0.854 \times 50\,\%$). Because an unregulated route carries a missing delay value rather than an explicit zero, the rule does not cover every comparison affected by a regulation. These results support learning from several inputs jointly.

Not all of those predictions carry the same weight. The model's confidence in each case is the gap between the scores it assigns to the two routes: a wide gap means a confident prediction, a narrow gap means the pair is close to a coin toss. As Section IV-B describes, a calibrator converts that gap into a probability, so that the model can say not only which route the airline would choose but how likely it is to be right. On the fifth of pairs where that confidence is highest, the model is right 94.9 % of the time. Fig. 3 traces the full trade-off: as the model abstains on low-confidence cases, the share of pairs it answers falls but its accuracy rises. This curve shows what is gained by withholding uncertain predictions, alongside the overall accuracy across all pairs.

Where those 68.1 % sit is more informative than the pooled figure, and Table IV splits the quarter two ways. Accuracy climbs steeply with the delay carried by the abandoned route, from 63.9 % where there was none to 84.5 % between half an hour and an hour, where the climb stops. This is consistent with stronger delay differences providing clearer information about the recorded choice, although the table does not isolate a causal effect of delay.

TABLE IV. ACCURACY BY STRATUM ON THE HELD-OUT QUARTER, WITH WILSON 95 % INTERVALS. THE TWO BLOCKS SPLIT THE SAME 281 720 PAIRS BY REGULATION EXPOSURE AND BY THE DELAY ATTRIBUTED TO THE ABANDONED ROUTE.

stratum	pairs	accuracy [%]
which route carried a regulation		
neither	178 370	64.0 [63.8, 64.2]
old route only	30 953	88.0 [87.6, 88.3]
both	67 587	70.6 [70.2, 70.9]
new route only	4 810	58.6 [57.2, 60.0]
delay on the abandoned route		
none	205 205	63.9 [63.7, 64.1]
up to 15 min	19 672	70.3 [69.7, 71.0]
15 to 30 min	18 283	79.3 [78.7, 79.9]
30 to 60 min	21 231	84.5 [84.0, 85.0]
above 60 min	17 329	83.5 [82.9, 84.0]

Accuracy ranges from 67.5 % to 69.0 % across April, May and June. This shows similar performance over the three test months; a longer follow-up would be needed to assess model ageing and retraining needs.

B. Change bias

Revision-only training sees how routes were changed, but not the alternatives considered when a route was retained. The stay-pair experiment tests whether adding examples of retention improves the model's predictions on that second kind of comparison.

A simple rule helps interpret this test. Always preferring the regulated route would be correct on every stay pair, because that is how the pairs are constructed. The same rule would be wrong on every revision that leaves a regulation. Performance must therefore be examined on both groups, not on stay pairs alone.

The effect is clear on both populations. Trained on revisions alone, the model scores 68.9 % on the revision pairs of the held-out quarter but only 34.8 % on the 1 093 stay pairs: it usually favours the alternative over the route that was retained. Adding stay pairs at group weight 10, meaning each stay pair counts ten times as heavily as one revision pair in the ranking loss, raises the stay-pair accuracy to 90.0 % and brings the revision accuracy to 68.1 %, a cost of 0.8 points. Table IV confirms that the adopted model also scores 88.0 % on flights that changed route after a regulation, the stratum where only the abandoned route carried a regulation, so the correction does not trade one population for the other. It rules out that fixed regulation-only rule, while leaving open whether other features reveal how the stay pairs were constructed. Training and testing stay pairs are from different periods; the test pairs were never added to the fitting set.

C. Error analysis and feature attribution

Errors concentrate where the model is already unsure. Counting the same high-confidence pairs by where their errors fall rather than by their accuracy: among the quarter of pairs whose two routes score furthest apart, only 5.6 % of all errors fall, against the 25 % expected if error were independent of confidence. In practical terms, most wrong predictions are cases the model was nearly indifferent about, and a policy

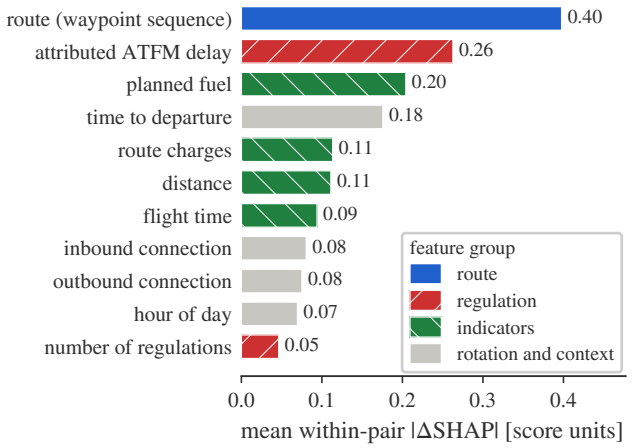


Figure 4. What moves a decision: the mean absolute *difference* in a feature's Shapley value between the two routes of a pair, over the held-out quarter, with the rotation and context groups of Table II drawn together. A larger bar means the feature more often shifts this model's comparison.

that abstains on those cases avoids most of its mistakes. The 10.0 % error rate on stay pairs measures how often the model favours the constructed alternative over the retained route. It is not a confident-error rate or a measured rejection rate: no confidence threshold is applied to that figure, and no actual proposal was sent.

To explain the model's scores, we compute Shapley feature attributions over the held-out quarter [23]. Each attribution assigns part of a route's score to an input feature. We then compare the feature's contribution on the two routes, because their score difference determines the predicted choice. For example, if a feature contributes $+0.3$ to one route and $+0.1$ to the other, it contributes 0.2 to their score difference. These illustrative values explain the calculation, not an additional result. Figure 4 averages the absolute size of such differences across the test quarter. A large bar means that the feature often influences this fitted model's comparisons; it is not an estimate of the airline's economic valuation of that feature.

On that scale the mean attribution difference is 0.40 for waypoint text, 0.26 for attributed ATFM delay and 0.20 for planned fuel. The waypoint text is the single largest term, ahead of every *individual* route indicator, though the four together exceed it, and attributed ATFM delay moves the score more than planned fuel does. The waypoint input therefore has a substantial role in the fitted model. The waypoint text thus carries information about the choice that the route indicators alone do not: the model's predictions depend on which route an airline files, as well as on the delay, fuel and charge differences studied here.

The waypoint text ranks first by attribution, yet removing it and retraining costs only 1.4 points of accuracy (68.2 against 66.8). An ablation removes an input, trains a new model and compares its accuracy with a model that retains that input. For instance, the 68.2 % and 66.8 % figures above come from a separate full-scale comparison; the first is not a different estimate of the main run's 68.1 %. This distinction matters when comparing small differences between runs.

The two numbers answer different questions: attribution measures how much the feature moves the score of the model that has it, while ablation measures how much worse a model trained without it would be. Where features are correlated, as route geometry and cost indicators plainly are, the model trained without the waypoint text recovers part of the lost information through the remaining features, so the ablation cost is smaller than the attribution rank would suggest.

D. Retrospective proposal evaluation

The proposal evaluation asks whether confidence filtering identifies lower-fuel routes that match recorded choices. Always selecting the lower-fuel route would match 44.9 % of eligible revisions. The model can improve on that baseline by abstaining when its predicted preference does not support the lower-fuel option.

Three measures must be kept separate. *Ranking accuracy* asks how often the higher-scoring route is the recorded replacement across test pairs. *Proposal precision* asks how often the selected lower-fuel route is the replacement, only among pairs passing the threshold. *Proposal coverage* is the fraction of eligible pairs that pass. The 94.9 % accuracy concerns the fifth of pairs with the largest score gaps, regardless of which route uses less fuel. Proposal precision instead concerns all lower-fuel candidates whose own probability exceeds the fixed threshold. A very confident prediction favouring higher fuel helps the first measure but produces no proposal under the second rule. Conversely, a less confident lower-fuel prediction can still pass the 0.600 threshold. The two measures therefore describe different subsets.

At the operating point selected to target 80 % precision ($\tau = 0.600$), the policy selects 56 162 pairs in the held-out quarter. Of these selections, 79.2 % match the later filing. For each match, we subtract the lower planned-fuel value from the higher one; incorrect selections contribute nothing to this sum. The total is 20.1 kt, equivalent to 63.5 kt of CO_2 at 3.16 kg per kilogramme of fuel.

Two annualisation assumptions scale the historical planned-fuel differences to 222 to 254 kt of CO_2; these are not forecasts of additional savings. Multiplying the quarter's total by four gives 254 kt. Applying the per-eligible-pair rate to an annual eligible volume estimated from the archive gives 222 kt.

These are sums of planned-fuel differences on matching historical revisions. They are neither measured fuel burn nor additional savings attributable to a proposal service: the airlines made those revisions without receiving proposals. The simulation also knows the replacement route retrospectively. A prospective evaluation must obtain feasible candidates and their estimated delays using only information available at the time of the proposed decision. The full proposal analysis is the subject of a journal extension of this work.

Figure 5 follows one of these proposals through the scoring pipeline. A narrow-body was filed on a medium-haul route with no regulation; the alternative saves 2 016 kg of planned



	abandoned	*filed*
planned fuel [kg]	10 485	8 469
flight time [min]	259	258
route charges [EUR]	2 269	2 496
model score	−2.82	+0.90

$$\text{gap} = 3.7 \longrightarrow P = 0.99 > \tau = 0.600 \longrightarrow \text{proposed}$$

Figure 5. A worked example from the held-out quarter. The route initially on file (dashed red) and the fuel-saving alternative (solid green) are shown on an anonymised city pair. The alternative saves 2 016 kg of planned fuel at 227 EUR more in route charges and the same flight time. The calibrated probability exceeds the threshold; the airline's own later filing chose the proposed route.

fuel at essentially the same flight time but 227 EUR more in route charges. The model scores the alternative 3.7 above the filed route, the calibrator maps that gap to a 99 % probability of matching the recorded choice, well above $\tau = 0.600$, so the policy proposes. The airline's later filing chose exactly that route. The margin is far from typical: the median confirmed proposal saves 130 kg, but this case shows each step of the retrospective evaluation. The archive does not provide the vertical profiles needed to explain the large planned-fuel difference at similar flight time.

VI. Conclusions

Flight-plan revisions provide useful training comparisons for predicting airline route choices. Both routes were filed for the same flight, so the label has a clear observational basis. With filing-time shortcuts removed, the model identifies the replacement route in 68.1 % of held-out revisions, 13.1 percentage points above the best tested single-indicator rule. On the fifth of pairs with the highest confidence, accuracy reaches 94.9 %. The result is therefore useful both as a ranking and as a way to identify more reliable predictions.

Adding stay pairs raises accuracy on that smaller test set from 34.8 % to 90.0 %, while revision accuracy changes from 68.9 % to 68.1 %. This demonstrates the importance of representing retained routes as well as revisions. The next data question is how to build stronger comparisons for flights that do not revise, with evidence that the alternative was available and regulation status does not identify the label.

The retrospective fuel-saving policy illustrates a separate use of the predictions. At a threshold selected to target 80 % precision, 79.2 % of selected routes match the subsequent filing, accounting for 20.1 kt of planned-fuel differences in the quarter. These figures measure selection among historical route pairs. They motivate research on proposal acceptance without estimating the additional fuel savings of a future service.

The two evaluations are complementary: revisions test which replacement is predicted, while stay pairs test recognition of retained routes. Neither population alone covers the full route-choice problem.

Route representation is another scientific question. Waypoint identifiers have the largest individual feature attribution, while removing them and retraining reduces accuracy by 1.4 points. The representation captures which published points a route uses, but discards their order and geometry. Initial experiments with a Transformer encoder of geometric route descriptors did not improve on the token-based model. Its embeddings also separated preferred from abandoned routes only slightly better than chance. Those findings motivate further work on representations that retain useful network context while depending less on waypoint names.

Finally, future evaluation should test how the model responds to changes in weather, operating conditions and fuel prices, and compare planned-fuel differences with flown trajectories. Prospectively recording candidate availability and airline responses would establish how well the predictable choices found here transfer to actual proposals.

Disclaimer

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

Use of artificial intelligence

In preparing this manuscript the authors used generative artificial-intelligence tools in two limited roles. First, an image model produced the initial artwork for the workflow diagram (Fig. 2); every label and value in that figure was then checked and corrected by hand against the method described herein and edited in a raster editor. Second, AI assistants helped with the language and clarity of the prose. The scientific content of the paper (the data, the model, the results and their interpretation, and the references) is the authors' own and was not generated by AI; every figure, number and citation was verified against the study's code and records.

REFERENCES

[1] A. Cook and G. Tanner, "European airline delay cost reference values: updated and extended values, version 4.1," University of Westminster, Tech. Rep., Dec. 2015, published by EUROCONTROL.

[2] P. A. Samuelson, "A note on the pure theory of consumer's behaviour," *Economica*, vol. 5, no. 17, pp. 61–71, 1938.

[3] R. Dalmau, P. Gascó, H. Kadour, É. Allard, and G. Gawinowski, "Learning to rank flight routes for improved air traffic demand predictions," in *Proc. 14th SESAR Innovation Days*, 2024, paper 095.

[4] D. Bertsimas and S. S. Patterson, "The traffic flow management rerouting problem in air traffic control: A dynamic network flow approach," *Transp. Sci.*, vol. 34, no. 3, pp. 239–255, 2000.

[5] A. Mukherjee and M. Hansen, "A dynamic rerouting model for air traffic flow management," *Transp. Res. Part B: Methodol.*, vol. 43, no. 1, pp. 159–171, 2009.

[6] L. Delgado, "European route choice determinants: examining fuel and route charge trade-offs," in *Proc. 11th USA/Europe ATM R&D Seminar*, Lisbon, Portugal, 2015.

[7] Z. Sha, K. A. Moolchandani, A. Maheshwari, J. Thekinen, J. Panchal, and D. A. DeLaurentis, "Modeling airline decisions on route planning using discrete choice models," in *15th AIAA Aviation Technology, Integration, and Operations Conf. (ATIO)*, 2015.

[8] M. C. R. Murça, "Collaborative air traffic flow management: Incorporating airline preferences in rerouting decisions," *J. Air Transp. Manag.*, vol. 71, pp. 97–107, 2018.

[9] W. Xu, M. Dalmau, D. Melgosa, A. Montlaur, and X. Prats, "A framework for collaborative air traffic flow management minimizing costs for airspace users: enabling trajectory options and flexible pre-tactical delay management," *Transp. Res. Part B: Methodol.*, 2020.

[10] T.-Y. Liu, "Learning to rank for information retrieval," *Found. Trends Inf. Retr.*, vol. 3, no. 3, pp. 225–331, 2009.

[11] M. Mateos Villar, I. Martín, P. Castillo, R. Herranz, O. García, and X. Prats Menéndez, "Full-scale pre-tactical route prediction: machine learning to increase pre-tactical demand forecast accuracy," in *Proc. Int. Conf. on Air Resource Management (ICRAT)*, 2020, openAlex W3171915093; UPC repository 2117/341212; no DOI.

[12] M. Mateos Villar, I. Martín, P. Castillo, R. Alcolea Rodríguez, R. Herranz, O. García, and X. Prats Menéndez, "Predicting requested flight levels with machine learning," in *Proc. SESAR Innovation Days (SIDs)*, 2020, openAlex W3157757530; UPC repository 2117/341211; no DOI.

[13] A. D. Evans, P. Lee, and B. Sridhar, "Predicting the operational acceptance of airborne flight reroute requests using data mining," *Transp. Res. Part C: Emerging Technol.*, vol. 96, pp. 270–289, 2018.

[14] A. Evans and P. Lee, "Using machine-learning to dynamically generate operationally acceptable strategic reroute options," NASA Ames Research Center, Tech. Rep. ARC-E-DAA-TN69401, 2019, nTRS 20190027644; conference paper.

[15] P. U. Lee, C. Brasil, C. Gabriel, K. Garzon, N. Buckley, and S. Kalush, "Demonstration of airline-based airborne reroute operations using trajectory option sets and third-party tools," in *AIAA AVIATION 2021 Forum*, 2021.

[16] S. Zelinski and M. Jastrzebski, "Defining dynamic route structure for airspace configuration," *Proc. IMechE Part G: J. Aerospace Eng.*, vol. 226, no. 9, pp. 1161–1170, 2011.

[17] C. Paradis and M. D. Davies, "Visualizing corridors in terminal airspace using trajectory clustering," in *Proc. 41st IEEE/AIAA Digital Avionics Systems Conf. (DASC)*, 2022, pp. 1–9.

[18] D. A. Karr, R. A. Vivona, and D. J. Wing, "Costs of limiting route optimization to published waypoints in the traffic aware planner," NASA Langley Research Center, Tech. Rep. NF1676L-15962, 2018, nTRS 20180001345; year inferred from accession number.

[19] C. Burges, T. Shaked, E. Renshaw, A. Lazier, M. Deeds, N. Hamilton, and G. Hullender, "Learning to rank using gradient descent," in *Proc. 22nd Int. Conf. Mach. Learn.*, 2005, pp. 89–96.

[20] L. Prokhorenkova, G. Gusev, A. Vorobev, A. V. Dorogush, and A. Gulin, "CatBoost: unbiased boosting with categorical features," in *Adv. Neural Inf. Process. Syst.*, 2018.

[21] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.*, 2002, pp. 694–699.

[22] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. 34th Int. Conf. Mach. Learn.*, 2017.

[23] S. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Adv. Neural Inf. Process. Syst.*, 2017.